Midterm Outline

GEOG 463 / 620

Outline

1. Format
2. Coverage
3. Review + Questions

Format

- **Multiple Choice**
  - About 45 questions: Concepts of GIS, **Situations**
- **Date/Time**
  - Oct 21st in class during lecture time 10:15 to 11:30
  - ACSD – accommodated according to the suggested time schedule received from ACSD, **should take midterm at the ACSD centre – Test should is booked already**

Coverage
Based on Lectures

- Week 1 – 5 Lectures & Readings

- Intro
- Map Projections
- Queries
- Geodatabases
- Network Analysis
- Raster Structure
- Decision Analysis
- Fuzzy Logic

Geodatabase: What is it, what can you do with it, why do we use it?

Queries: what are they? Can you find the result of a query? Can you convert a general question into a query? If you see a result/subset, can you find out what the query was?

Table

Mtl_CT_Census2006_WGS84

	SRID	SRID_1	Pop01	Pop06	MeanChild	Own
▶	0001.00	0001.00	2445	2498	0.9	520
	0002.00	0002.00	2734	2819	0.9	555
	0003.00	0003.00	6522	6407	1	1230
	0004.00	0004.00	3213	3223	0.9	620
	0005.00	0005.00	3171	3188	0.8	535
	0006.00	0006.00	4541	4568	1.1	485
	0007.00	0007.00	5699	5712	0.8	1420
	0008.00	0008.00	2957	2973	1	750
	0009.00	0009.00	3657	3554	0.9	660
	0010.00	0010.00	1444	1351	0.8	280
	0011.00	0011.00	4652	5160	0.9	1000
	0012.00	0012.00	0	0	0	0
	0013.00	0013.00	4355	4427	0.8	850
	0014.00	0014.00	0	0	0	0
	0015.00	0015.00	3421	3522	1	195
	0100.00	0100.00	7483	7618	1.1	1090

1 (0 out of 913 Selected)

Mtl_CT_Census2006_WGS84

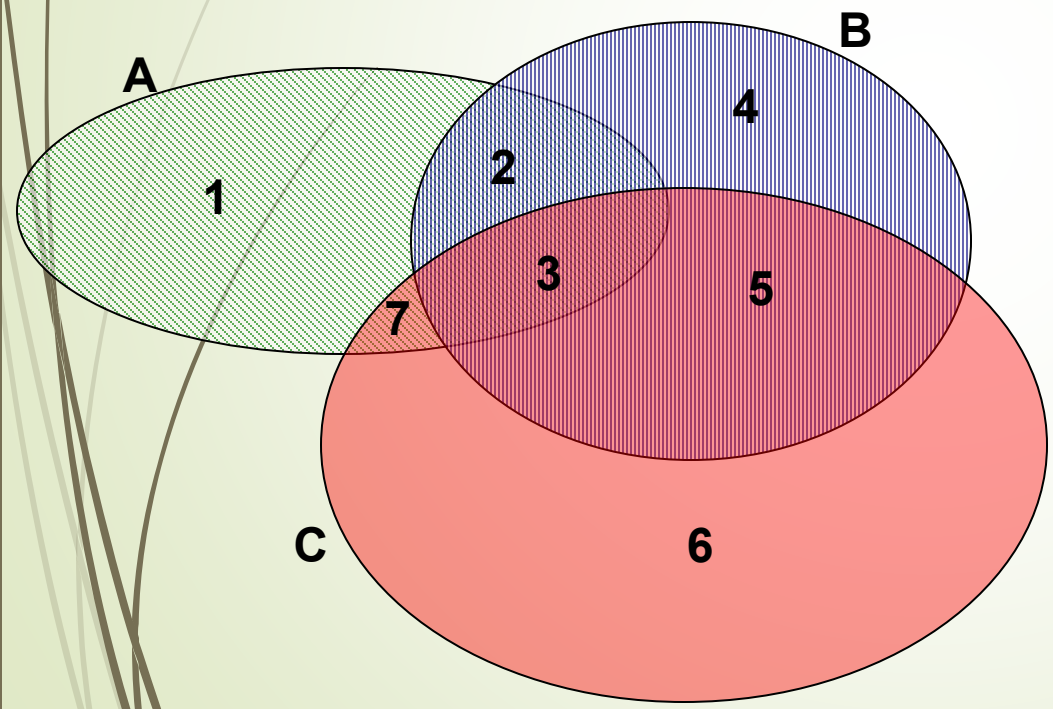
Find the census tracts where the 2001 population is greater than the 2006 population; and where the average number of children was greater or equal to 1

Select * *From*
<database> *Where*

What does the following query return?
Select * from <table> WHERE ("Pop06"> 5000 AND "MeanChild >1) OR ("Pop06 > Pop01)

SQL: Query building

Queries are evaluated from according to standard operator precedence rules. **Expression enclosed in parentheses** is evaluated before the part that isn't enclosed.

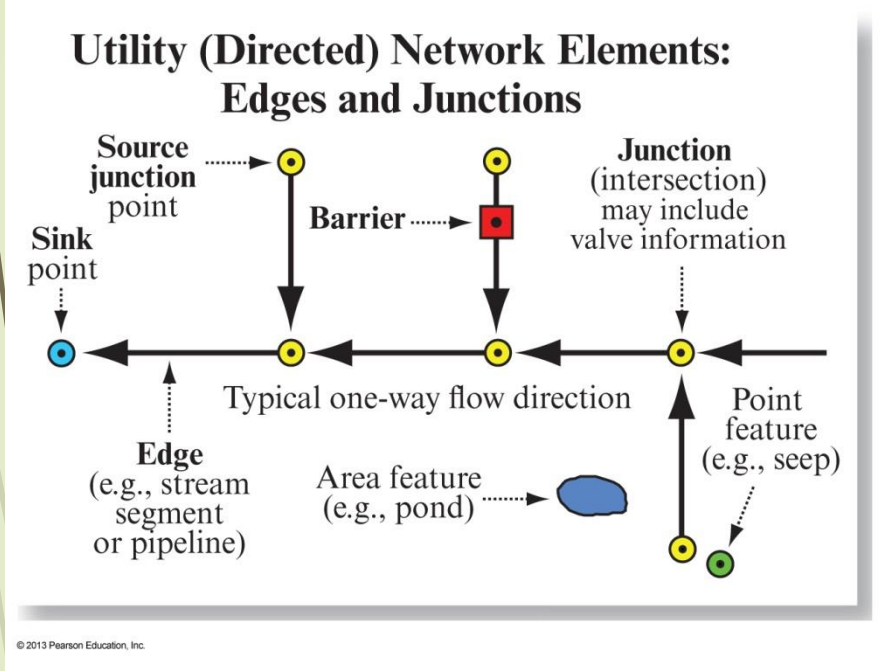


Is
A AND B OR C
the same as
A AND (B OR C)?

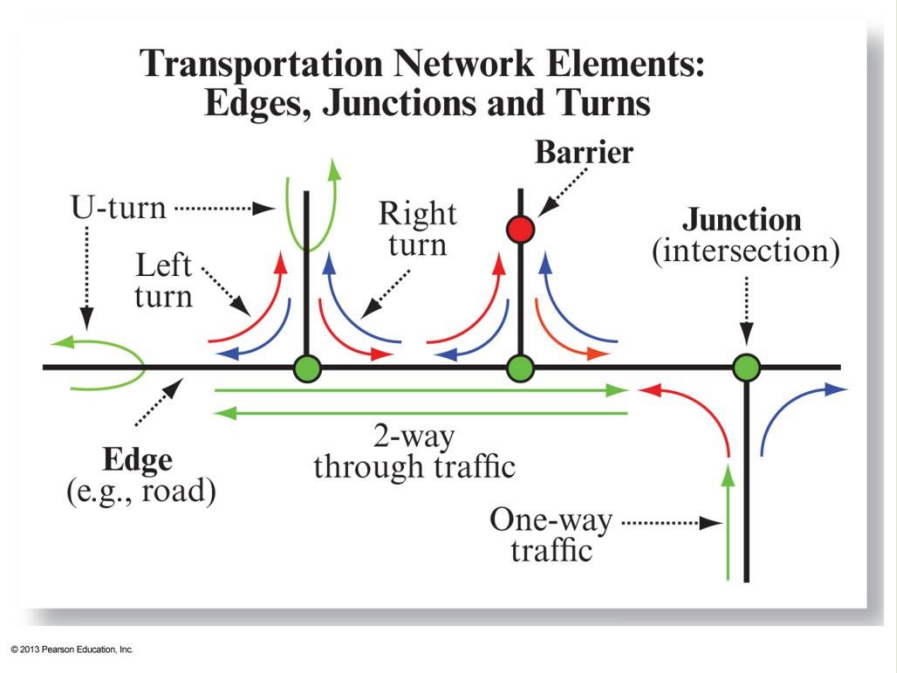
A AND B OR C ➔ 2, 3, 5, 6, 7
A AND (B OR C) ➔ 2, 3, 7

Network Analysis: what is it, which types?

UTILITY



TRANSPORTATION



Jensen and Jensen (2013)

Coordinates/Projections

General understanding, Situations / Cases

Data Storage: what is it, why do we do/need it, which types?

Decision Strategy Analysis: what is it, why, how do we do it? Situations…

Fuzzy analysis: what is it, why, how?

1. State the problem / objective

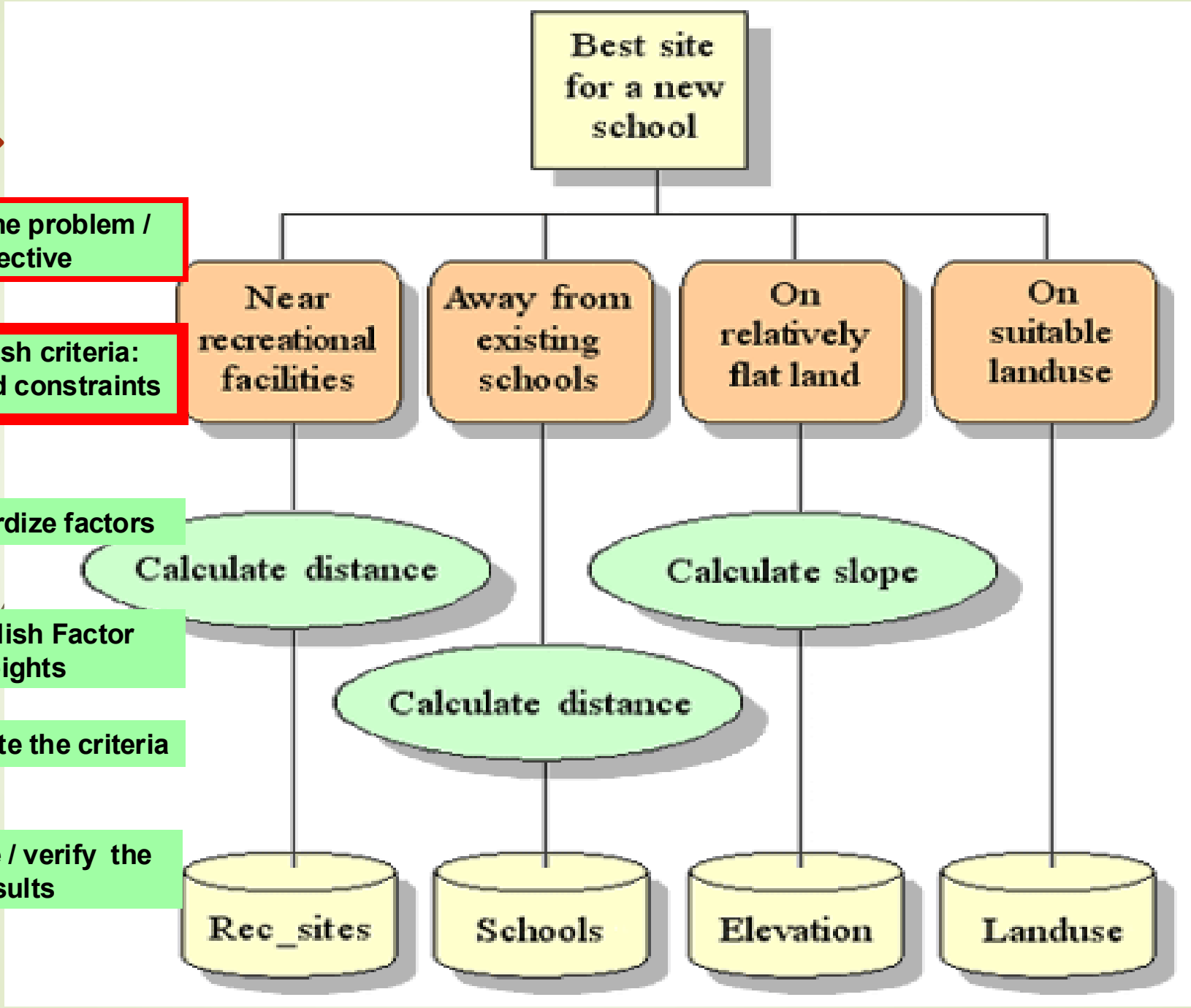
2. Establish criteria: factors and constraints

3. Standardize factors

4. Establish Factor Weights

5. Aggregate the criteria

6. Validate / verify the results



Data Storage: what is it, why do we do/need it, which types?

Decision Strategy Analysis: what is it, why, how do we do it? Situations…

Fuzzy analysis: what is it, why, how?

1. State the problem / objective

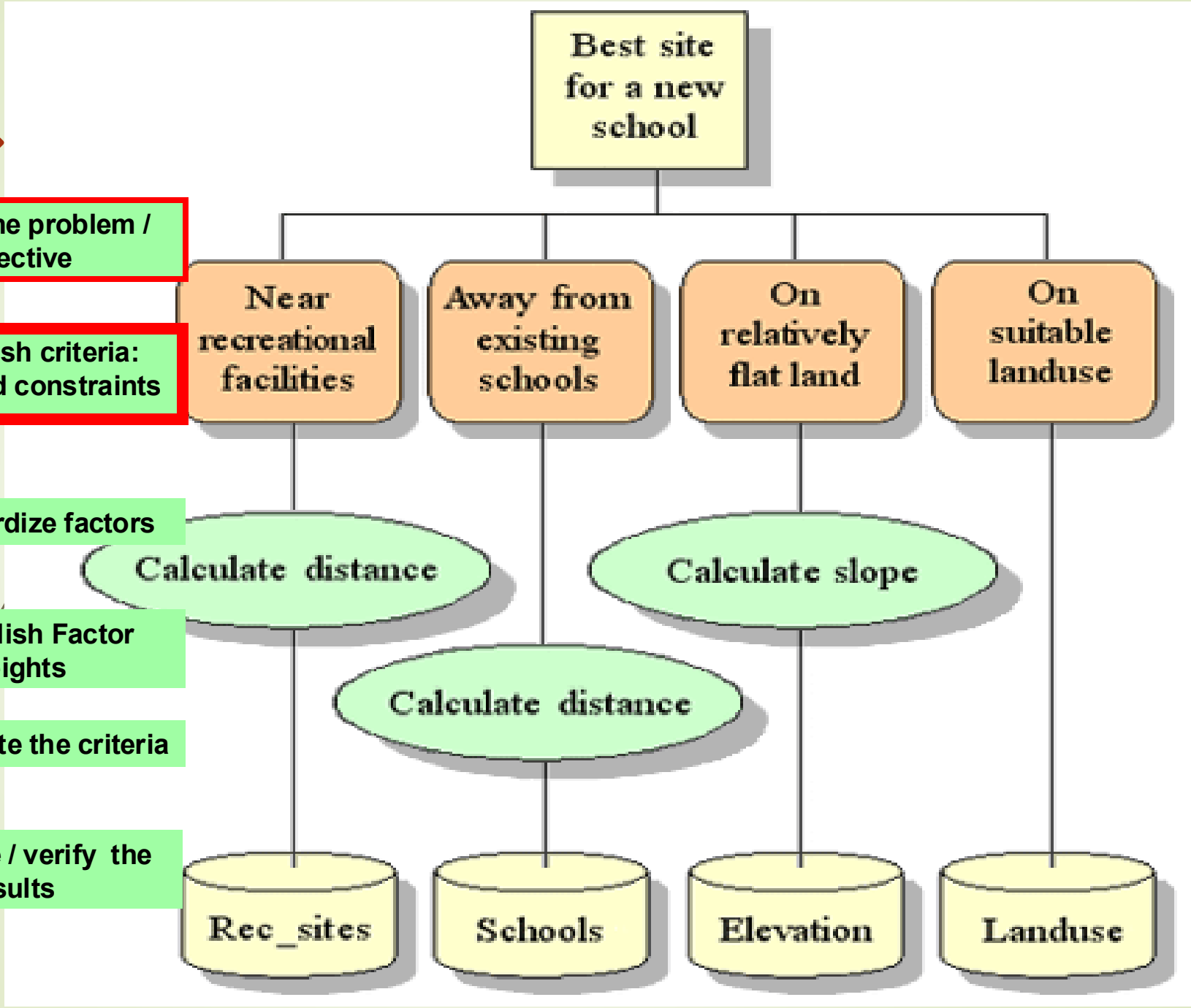
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Exercise
Site selection for new housing development

Town regulations require residential development to be at least 100 m from open water and wetlands. Suitability very close to the protected zone should therefore remain very low. However, at distances greater than 800 m, suitability should be considered as very high.

1) Draw the fuzzy function (linear member ship function);
   - Is it increasing or decreasing?
   - What are the values of the breakpoints *a* and *b (or c and d)*?
   - What is the function?



You found a great site at a distance of 600 m. The site selection committee determined that sites with a membership of 60% (or 0.6) or greater will be considered suitable for construction. Will this great site that you found be suitable?

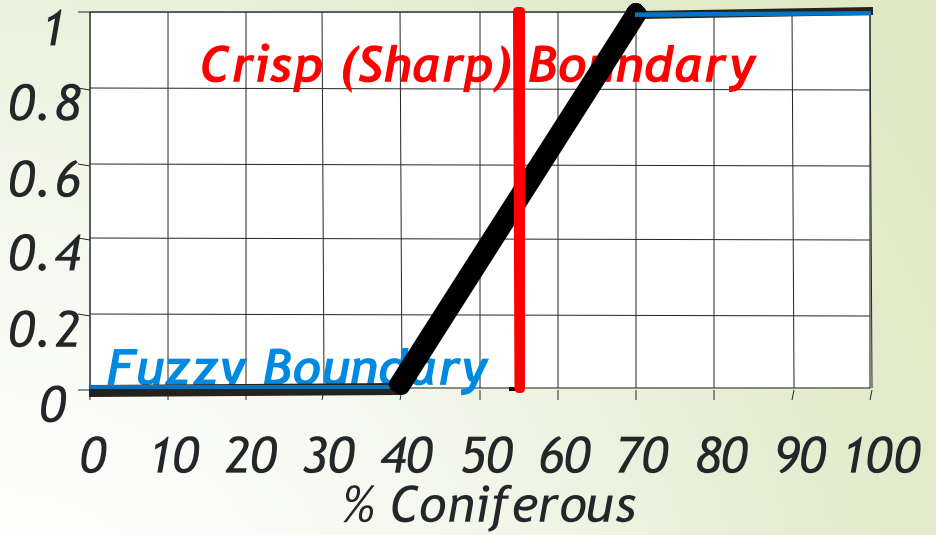
Increasing linear function: μ = (x – point a) / (point b – point a) * range

Decreasing linear function: μ = {1 – [(x – point c) / (point d – point c)]} * range

Example - Coniferous Forest

$$\mu(x) = \frac{x - 40}{70 - 40}$$

$$\mu(x) = \frac{x - a}{b - a}$$



Fuzzy - not coniferous

$\mu(x) = x \leq 40$

Forest has less than or equal to 40% coniferous trees, so it is not coniferous

Fuzzy - transition

Forest consists of between 40% and 70% coniferous trees so we compute $\mu(x)$.

Fuzzy - coniferous

$\mu(x)=1 \quad x \geq 70$

Coniferous trees compose more than or equal to 70% of the forest so it is a coniferous forest.